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A Causal Evaluation of Timeouts in the NBA

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Abstract

There is a long-standing belief in the NBA that calling a timeout during an opponent's scoring run can disrupt their momentum and improve the calling team's chances of winning. However, the empirical evidence in support of this belief is limited and inconclusive. In this paper, we use play-by-play data from the 2021-2026 NBA seasons to conduct a causal analysis of the effect of timeouts on team performance.

Keywords: Causal inference, sports analytics, NBA, timeouts, momentum, difference-in-differences, synthetic control

1 Significance

The definition and nature of streaks, runs, and momentum itself has been a point of interest and debate for sports fans and analysts for decades. In addition to the papers cited in our related works, there's a large body of research on the existence and nature of momentum in sports, both on the individual level (e.g. the "hot-hand" effect) and the team level.

On an academic level, the question of whether timeouts disrupt momentum is a interesting case study in causal inference in a complex, dynamic system. It involves issues of selection bias (coaches call timeouts in response to game state), confounding (many factors influence scoring dynamics), and the challenge of defining and measuring momentum itself. The conflicting conclusions in the existing literature suggest that this is a non-trivial question.

The significance of this question extends beyond academic curiosity. If timeouts do have a causal effect on disrupting momentum, it would validate a common coaching strategy and could influence how teams manage their timeouts. There's also opportunity for further discovery: how does the relationship change conditioned on player substitutions, home-away status, quarter, regular vs post season, team age and maturity, duration and intensity of the run, etc.? There are many more such ablations to consider. Conversely, if timeouts are found to have no effect or even a negative effect, it could lead to a reevaluation of how coaches use their timeouts and potentially change the way teams approach in-game strategy.

2 Related Works

The question of whether timeouts actually disrupt momentum has been studied from several angles, with notably conflicting conclusions. Assis et al. (2020) apply a formal causal framework to NBA play-by-play data, using matching techniques informed by a causal graph to estimate the counterfactual effect of a timeout. They conclude that timeouts have *no* effect on team performance, and argue that the apparent rebound observed after a timeout is simply regression to the mean: because timeouts tend to be called when the opposing team is scoring at an unusually high rate, any subsequent slowdown reflects the natural tendency of scoring to return to its average rather than a causal effect of the stoppage itself.

Gibbs et al. (2021) revisit the same question but restrict attention to the specific scenario of an opposing team's scoring run. Working within the Rubin causal model, they carefully define units under SUTVA and introduce an interpretable outcome based on the score difference to capture broad changes in scoring dynamics. In contrast to Assis et al. (2020), they find that calling a timeout during an opposing run is *slightly disadvantageous* on average, and further show that the magnitude of this effect varies meaningfully by franchise. Their result suggests that comebacks after runs happen frequently enough to create the illusion of timeout efficacy, but that the intervention itself does not help.

Weimer et al. (2023) take a different identification strategy by exploiting mandatory television timeouts, which are triggered at predetermined points in the game and are therefore plausibly exogenous with respect to in-game momentum. Using TV timeouts as a natural experiment that mimics random assignment of a pause, they report an *11.2% decline* in the scoring rate of the team carrying momentum immediately after a TV timeout, and they show this effect is robust to run size, substitutions, and game context. This stands in sharp contrast to the null result of Assis et al. (2020) and the mildly negative result of Gibbs et al. (2021); by removing coach selection bias entirely, Weimer et al. (2023) argue that team-level momentum genuinely exists and that pauses in play do interrupt it.

3 Methods

3.1 Data

We build our analysis on the cdnnba enriched play-by-play data for the 2020–2025 NBA seasons (regular season and playoffs). After cleaning, the dataset contains 4,168,786 play-by-play rows spanning 7,457 unique games and 1,480,697 possessions, of which 82,197 are coach-logged timeouts.

3.2 Mandatory Timeout Rule Changes

For to the 2017-18 season, the NBA Board of Governors overhauled the timeout format (National Basketball Association, 2017; NBA.com Staff, 2017). Moving forward, there would be 2 mandatory TV timeouts per period, triggered at the 6:59 and 2:59 marks. In addition, they changed the rules to deduct the first unused mandatory timeout from the home team, and the second from the visiting team. *Example*: If neither team has called a timeout by the 6:59 mark, the official scorer calls one and charges it to the home team; if no subsequent timeout is taken before the 2:59 mark, the scorer calls one and charges it to the visiting team.

This rulebook change has two implications for our analysis. First, it changes the settings of our analysis from those conducted by Weimer et al. (2023), which used data from 2013–

2017. Secondly, it creates a new category of *pseudo-mandatory* timeouts: coach calls that occur in the seconds leading up to their imminent mandatory trigger.

3.3 Timeout classification

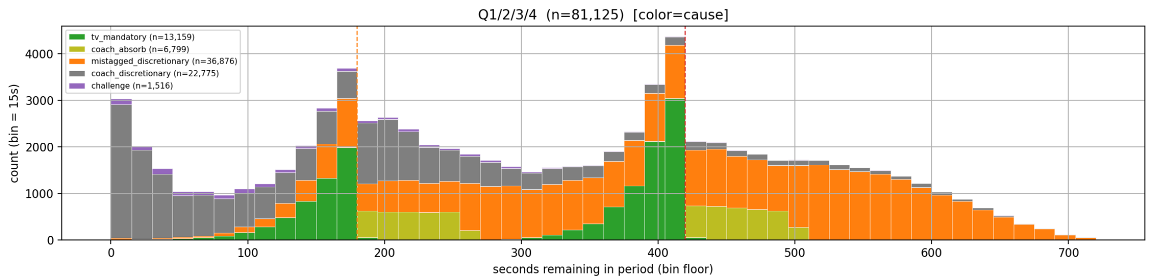

Figure 1: Distribution of cdnnba timeouts by cause and by seconds remaining in the period, all four regular periods combined. The two dashed vertical lines mark the rulebook trigger points (6:59 and 2:59).

Of those 82,197 timeouts, the composition is as follows:

- 13,172 `tv_mandatory` rows (league-forced commercial breaks at roughly the 6:59 and 2:59 marks of each period). These serve as our primary 'exogenous' treatment group.
- 6,805 `coach_absorb` rows. These are coach called timeouts that occur right before (within 80 seconds) a mandatory timeout would have triggered.
- 23,747 `coach_discretionary` rows. These are coach called timeouts
- 36,918 `mistagged_discretionary` rows. These are coach called timeouts that were tagged as "mandatory" by the feed, but **we reclassified** as discretionary in nature.
- 1,555 `coach_challenge` rows. These are coach challenges, which are a special type of timeout that allow for replay review and potential overturning of the preceding call.

Endogenous events are the union of the coach_discretionary and mistagged_discretionary categories, which are both coach-initiated. While coach_challenge events are also coach-initiated, we keep them separate as a secondary treatment group because they have an additional mechanism (replay review) and a much smaller sample size.

Exogenous events are the union of the tv_mandatory and coach_absorb categories together with natural game stoppages (out-of-bounds, injuries, blood rule, equipment). All three share the same "pause in play" mechanism with no coach selection on game state; only the trigger differs.

Control events are *play-continuing* moments during an active scoring run – rows where neither a timeout nor a stoppage occurred, so play proceeded normally. We sample at most one control per run segment to avoid over-representing long runs.

NBA rules allow substitutions and strategy huddles during *any* timeout (full, short, mandatory TV, challenge, injury), so the opportunity to alter play after the break is identical in all cases. What separates the two types is how the trigger is selected: coaches initiate endogenous timeouts in response to game state (introducing selection bias on adverse momentum), whereas mandatory TV timeouts and coach_absorb events fire at fixed clock positions and are therefore exogenous conditional on the clock.

3.4 Run and recovery definitions

A *scoring run* at play t is defined as $|\texttt{streak}(t)| \geq s$, where $\texttt{streak}(t)$ is the cumulative net score swing within the current scoring segment, computed by resetting the counter whenever the scoring team flips. We refer to the team being outscored as the *suffering team* and the team scoring as the *running team* or alternatively the *streaking team*.

Our primary outcome is the *recovery* metric: the signed lead change from the suffering team's perspective over a forward window of n minutes. If $\texttt{lead}(t)$ is the home-minus-away margin at play t, we define

$$\texttt{recovery}(t, n) = -\operatorname{sign}(\texttt{streak}(t)) \cdot \big(\texttt{lead}(t + n \text{ min}) - \texttt{lead}(t)\big),$$

so that positive values mean the suffering team clawed points back during the window. As a secondary outcome we also report `running_team_pts`, the raw points scored by the running team over the forward window, which matches the dependent variable used by Weimer et al. (2023).

3.5 Between-group estimators

As a first pass we compute group means of `recovery` conditional on run size s and forward window n (measured in minutes), and report differences versus the control group using Welch's t-tests. We sweep $s \in \{4, 5, 6, 7, 8, 10, 12, 15\}$ and $n \in \{0.5, 1, 1.5, 2, 3, 4, 5\}$ minutes, and also stratify by quarter, regular season vs. playoffs, and absolute score margin.

3.6 Within-game matched-twin causal design

Motivation. The between-group estimator is vulnerable to game-level confounding: coaches call timeouts in systematically different games than the ones where control events are drawn, and mandatory TV timeouts cluster at predictable game-clock positions (around the 6:59 and 2:59 marks of each period) rather than uniformly across all game contexts. Comparing the timeout pool against a global control mean therefore absorbs game- and clock-level selection into the baseline. To remove this bias we use a *within-game matched-twin* design that holds team composition, pace, coaching, and game-level momentum approximately fixed by construction.

Method. We first build a candidate pool by classifying every spine event during a run of $|\texttt{streak}| \geq s$ into one of (**Endogenous**, **Exogenous**, **Control**). For each treated event we then search the control pool *within the same game and period* for a candidate that satisfies all of: the same sign of `streak` (so the run is against the same team); $|\Delta\,\texttt{streak}| \leq 2$ (comparable run magnitude); $|\Delta\,\texttt{seconds\_remaining}| \leq 120\,\text{s}$ (comparable clock position within the period); and $|\Delta\,\texttt{suffering\_margin}| \leq 3$, where the suffering margin

$$\texttt{suffering\_margin}(t) \;=\; -\,\mathrm{sign}(\texttt{streak}(t)) \cdot \texttt{lead}(t)$$

is the score margin from the suffering team's perspective (positive when the suffering team is still ahead despite the run against it). When multiple candidate controls satisfy the constraints, we select the one minimizing the weighted Manhattan distance

$$d(\text{treated, control}) \;=\; 2\,|\Delta\,\texttt{streak}| + \frac{|\Delta\,\texttt{seconds\_remaining}|}{30} + 3\,|\Delta\,\texttt{suffering\_margin}|,$$

which trades a ~30-second clock gap for one point of streak magnitude or one point of score margin. Each treated event yields at most one pair; the same control row may serve as the matched twin for multiple treated events. The per-pair causal estimate is the difference in `recovery` between the treated event and its matched twin, and we aggregate across pairs with a paired t-test.

3.7 Moderator analyses

To probe heterogeneity we repeat the between-group and matched-twin analyses within slices defined by (i) timeout subtype (`tv_mandatory`, `coach_absorb`, `coach_discretionary`, `mistagged_discretionary`, `coach_challenge`, `stoppage`); (ii) game phase and clutch status (last five minutes of Q4 with margin ≤ 5); (iii) season-long team quality gap, using each team's `NET_RATING` from `player_advanced_stats`; (iv) the number of substitutions within a $\pm$30s game-clock window around the event, bucketed as $\{0, 1\text{–}2, 3+\}$; and (v) the running team's in-game cumulative points-per-possession (PPP) at the time of the event, as a proxy for short-term shooting rhythm.

4 Evaluation

4.1 Between-group results

The simple between-group comparison essentially reproduces the weak-to-null effects reported in the prior literature. Across run sizes $s \in \{5, \ldots, 10\}$ and forward windows $n \in \{1, \ldots, 5\}$ minutes, the endogenous-vs-control gap in `recovery` is never statistically significant (all $|\Delta| < 0.1$ pts, $p > 0.05$ in 15 of 16 cells). Exogenous timeouts, by contrast, look significantly *negative*, with $\Delta \approx -0.25$ to -0.45 pts at the 3-minute window depending on run size ($p < 0.001$). Sweeping the forward window at fixed run size $s = 6$ shows the exogenous penalty growing from -0.20 at 30 seconds to a plateau around -0.27 at 3 minutes, while the endogenous gap hovers near zero throughout. At face value this matches the "timeouts don't help (and mandatory TV breaks hurt)" consensus. As a secondary check we also computed `running_team_pts` in the style of Weimer et al. (2023) and found a *positive* exogenous-minus-control gap of +2.8% to +14.6% depending on window – the opposite sign from Weimer's reported -11.2%. This is internally consistent with our suffering-team analysis (if the suffering team recovers less, the running team must score relatively more) and likely reflects the post-2017 mandatory-timeout rule changes together with differences in how we classify mandatory timeouts.

To avoid confusion we restate the result: under a between-group estimator, endogenous coach timeouts look indistinguishable from control, and exogenous breaks look mildly negative. We believe this is **not** a causal effect but instead a consequence of the adverse situations in which coaches predominantly call timeouts – a game-level selection bias that is absorbed into the global control baseline. The matched-twin design of §4.2, which holds game identity (and therefore that selection) fixed by construction, will reverse this conclusion and produce a strongly positive and *symmetric* causal estimate for both endogenous and exogenous breaks.

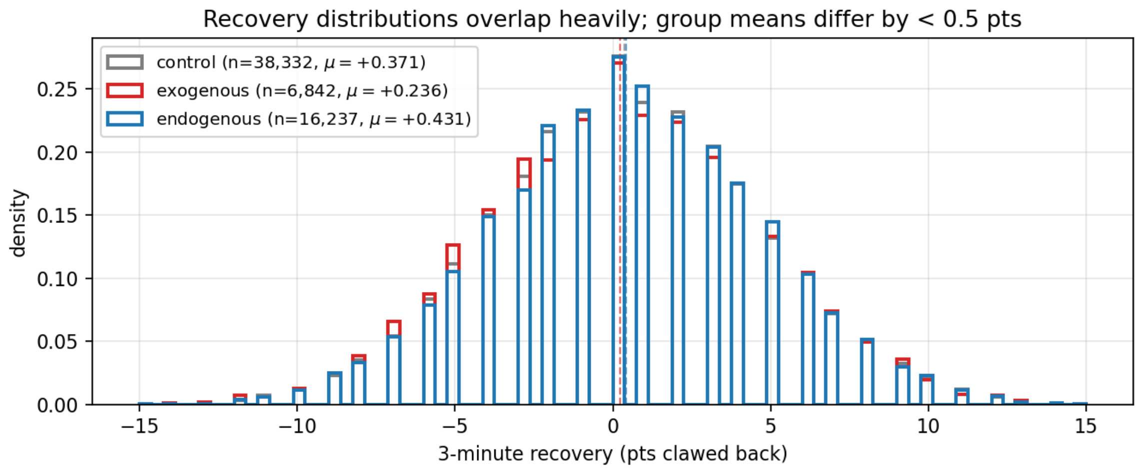

Figure 2: Distribution of 3-minute `recovery` across the three groups. The three densities overlap almost entirely; the group means (dashed verticals) differ by less than half a point.

4.2 Within-game matched-twin results

The matched-twin design of §3.6 – our cleanest causal test – reverses the headline conclusion of the between-group analysis. With $s = 6$ and a 3-minute window we obtain 7,474 matched endogenous pairs and 3,480 matched exogenous pairs (each treated event matched to its closest in-game control by the weighted distance d; control rows are reused across treated rows when they are the best match for multiple). The paired mean differences are +0.408 **pts for endogenous timeouts** ($t = 26.5$, $p < 0.0001$) and +0.405 **pts for exogenous timeouts** ($t = 13.2$, $p < 0.0001$). Note that these values are extremely similar. Breaking the treated population into the six fine subtypes of §3.3 reveals the same sign and roughly the same magnitude everywhere: `tv_mandatory` +0.361 *** ($n = 879$), generic `stoppage` +0.458 *** ($n = 1{,}752$), `coach_absorb` +0.340 *** ($n = 849$), `coach_discretionary` +0.429 *** ($n = 2{,}119$), `mistagged_discretionary` +0.400 *** ($n = 5{,}355$), and `coach_challenge` +0.469 ** ($n = 98$). The consistency across six structurally distinct event types – league-forced auto- fires, natural stoppages, last-second coach absorbs, pure discretionary calls, mistagged-as-mandatory coach calls, and replay challenges – is itself evidence that the ef-

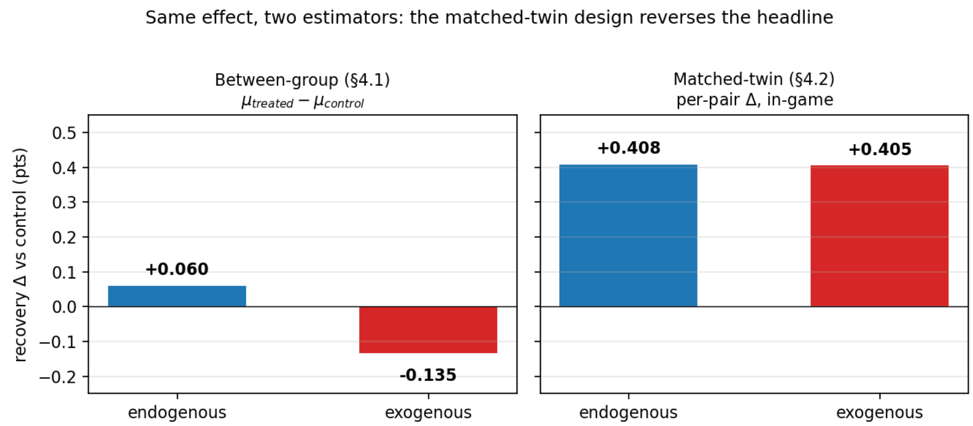

Figure 3: The same effect under the two estimators; The shift is not from finding a new effect; it is from differencing out the game-level selection bias that contaminated the between-group baseline.

fect is driven by the pause itself, not by who chose to take it or how the league billed it.

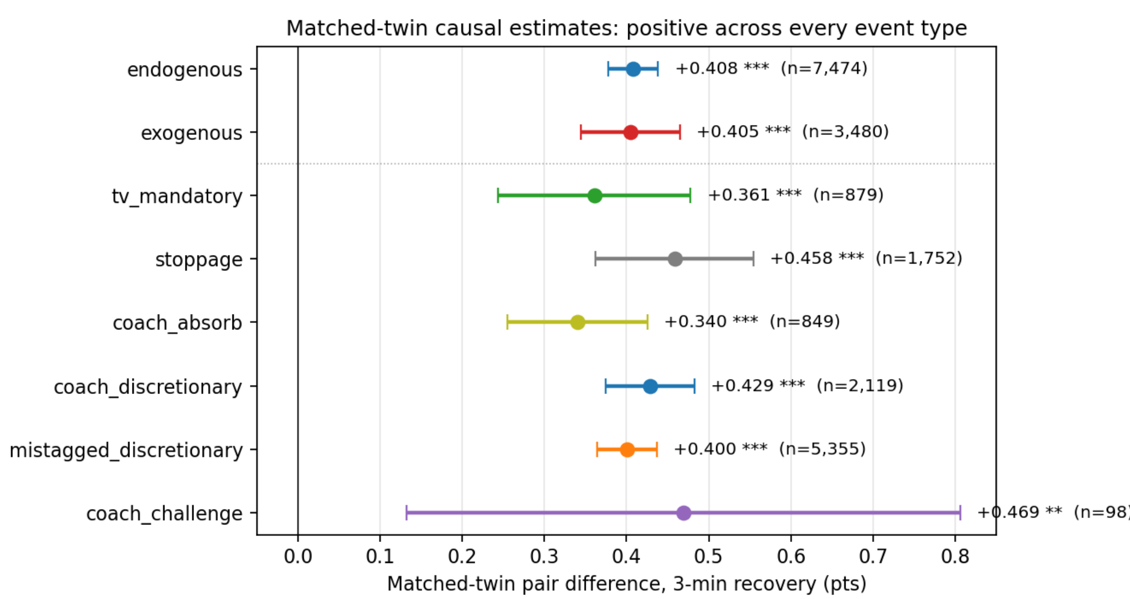

Figure 4: Forest plot of matched-twin pair differences with 95% confidence intervals. The top two rows are the coarse **endogenous** and **exogenous** aggregates; the six rows below are the fine subtypes from §3.3.

The reversal relative to the between-group results is driven by game-level selection: coach timeouts concentrate in games where the suffering team historically claws back less of its deficit during opponent runs (low-recovery games), so comparing the endogenous mean to a global control mean absorbs that selection into the baseline and erases the signal. The matched twin lives in the same game, same period, with the same streak sign and a near-identical clock position and score margin, so any game-level confounder (team quality, pace, coaching style, overall momentum direction) is differenced out at the pair level and the estimator survives whatever selection-on-game produced the negative exogenous gap in §4.1.

4.3 Mechanism and moderators

Once we have a signal, we can ask where it comes from. To that effect, we conducted Welch two-sample t-tests, whose values we report below – that is, $\Delta = \mu_{\text{treated}} - \mu_{\text{control}}$, the difference between the mean `recovery` of treated events and the mean `recovery` of control events within each stratum.

Stratifying by the number of substitutions in a $\pm$30s window around the event reveals that the endogenous effect is concentrated entirely in timeouts accompanied by personnel changes: with 0 substitutions the endogenous-control gap is -0.06 (n.s.), with 1–2 substitutions it is $+0.11$ (n.s.), and with 3 or more substitutions it rises to $+0.22$ ($p < 0.001$). The exogenous penalty is correspondingly largest (-0.22, $p < 0.05$) when no subs occur and vanishes when many do. This directly confirms the mechanism proposed by Weimer et al. (2023): the value of a timeout is mediated by substitutions rather than by the pause itself.

Conditioning on team quality via season-long `NET_RATING`, we find that the endogenous benefit is strongest in evenly matched games ($+0.32$ pts, $p < 0.01$) and washes out in lopsided matchups where the talent gap dominates the trajectory. Finally, stratifying by the running team's in-game PPP we find that when the opponent is shooting unusually efficiently (PPP $\geq$ 1.30), the two effects diverge in sign: endogenous timeouts produce $+0.20$ pts of extra recovery (n.s. at n = 1,595) while exogenous timeouts produce -0.37 pts ($p < 0.05$). The asymmetry is extremely suggestive of a strategic coaching response to hot-shooting opponents, and it matches the conditional-momentum findings of Qiu et al. (2025) that the value of a timeout is largest when the opponent is on a hot streak.

4.4 Summary

Our headline finding is that timeouts causally improve the suffering team's short-term recovery by roughly $+0.4$ points over the following three minutes, and that this effect is masked in between-group analyses by a subtle but severe game-level selection bias. The within-game matched-twin estimator, which holds game identity approximately fixed by construction, produces a strongly positive and highly significant effect of $+0.408$ **pts** for endogenous (strategic coach) breaks and $+0.405$ **pts** for exogenous (league-forced or auto-fire) breaks – two estimates that are statistically indistinguishable. That symmetry, together with the consistent $+0.34$ to $+0.46$ pair-difference across six structurally distinct fine subtypes, is strong evidence that the effect is driven by the pause itself rather than by who chose to take it. The published consensus of "no effect" or "slightly negative effect" (Assis et al., 2020; Gibbs et al., 2021) reproduces in our data only under a between-group estimator that

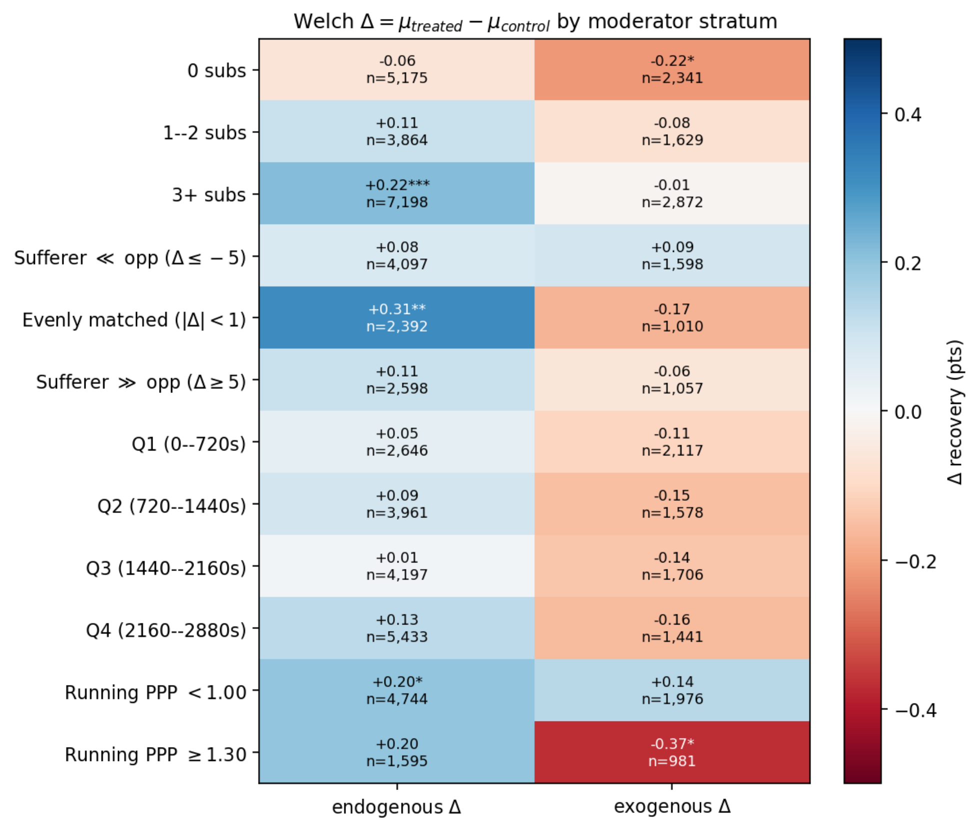

Figure 5: Welch $\Delta = \mu_{\text{treated}} - \mu_{\text{control}}$ across moderator strata. Blue cells favor the suffering team; red cells suppress recovery. The endogenous column is most positive when many substitutions occur ($+0.22$ $p < 0.001$) and in evenly matched games ($+0.31$ $p < 0.01$).

absorbs the game-level selection into the baseline. The benefit is overwhelmingly mediated by substitutions (in agreement with Weimer et al. (2023)) and is largest in evenly matched games and against hot-shooting opponents, which matches both coaching intuition and the conditional-momentum findings of Qiu et al. (2025).

5 Future Work

Several extensions follow naturally from the present design:

- **Better pairing mechanism.** How should we score the quality of a matched twin? The current Manhattan-weighted distance d is a reasonable first cut, but there is a sweet-spot to characterize between pair similarity (which tightens the estimate) and matched-pair sample size (which determines its precision).

  In addition, the current design nearly guarantees that the two matched events overlap in time, reducing the experiment to a test of the immediate effect of substitutions rather than the broader effect of the timeout itself. A redesigned match that relaxes the per-period clock constraint and allows for non-overlapping pairs would enable us to test the substitution mechanism more directly by comparing the effect size of overlapping vs. non-overlapping pairs.

- **Alternative outcome metrics.** Beyond `recovery`, we could measure the streaking team's points-per-possession or points-per-minute over the forward window, as was done in Weimer et al. (2023), or roll either up into a win-probability impact via a calibrated WP model.

- **Causal graphical model.** Distil the moderator findings into an explicit CGM (substitutions → recovery; running-team PPP → recovery; `NET_RATING` gap → recovery) and use it to formalize the identification arguments and check for missing confounders.

- **Streaking-team timeouts.** Re-run the analysis with the *streaking* team as the calling team. Do they substitute players at a higher rate than baseline, and what mechanism would explain a positive or negative effect on the run they are already riding?

- **Generic Stoppage Evaluation.** It's interesting that the generic `stoppage` category has a much higher point estimate (+0.458) than the league-forced `tv_mandatory` category (+0.361). Could this be noise, or is there another causal mechanism at play here?

5.1 Minor extensions

- *(open)* Analyse real-world elapsed time as a mechanism: do longer breaks correlate with larger recovery?

- *(open)* Team age as a moderator. Use the average age of players on the court at the time of the break, plus the delta in average age before vs after the break (a proxy for whether the coach swapped out tired veterans for fresh legs).

- *(open)* End-of-quarter trimming: never include events in the last k (say, $k = 2$) minutes of a period, and only include forward-window outcomes that fit inside the period (e.g. with 3.5 minutes left, accept the 3-minute window but exclude the 4-minute one). Would tighten the controls but cost sample size.

- *(open)* Investigate simple substitutions without official timeouts (e.g. at free throws). When else are teams allowed to substitute? A clean substitution-without-pause control would isolate the personnel-change mechanism from the pause itself.

- *(open)* Re-run with the forward window measured in *possessions* instead of minutes, ensuring the window contains an even number of possessions so each team gets the same opportunity count (modulo the rare technical-foul edge case).

Acknowledgments and Disclosure of Funding

N/A

LLM Usage

`claudeopus-4.6` was utilized extensively during the implementation of this research project, and we wanted to acknowledge its contributions.

The author(s) were responsible for project idea generation, sourcing relevant papers and publications, research roadmap, code scaffolding, and report skeleton.

`Opus` assisted in the implementation: data preprocessing and cleaning, feature generation, significance testing, ablation studies, and report completion. All `Opus` work was closely reviewed and cross-examined.

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Appendix

Tables below reproduce the post-rerun E9–E14 results from the analysis pipeline driver (`run_experiments_v2.py`) on the cdnnba dataset, run size $s = 6$, 3-minute forward window, with the cause-based reclassifier and strict slot-owner gate of §3.3. Significance codes: n.s. $= p \geq 0.05$, $* = p < 0.05$, $** = p < 0.01$, $*** = p < 0.001$.

E9: Matched-twin causal estimates

group	pairs	treated μ	matched ctrl μ	pair diff	sig
endogenous	7,474	+0.350	−0.058	+0.408	***
exogenous	3,480	+0.198	−0.207	+0.405	***

subtype	pairs	treated μ	matched ctrl μ	pair diff	sig
tv_mandatory	879	+0.259	−0.101	+0.361	***
stoppage	1,752	+0.059	−0.399	+0.458	***
coach_absorb	849	+0.422	+0.081	+0.340	***
coach_discretionary	2,119	+0.352	−0.076	+0.429	***
mistagged_discretionary	5,355	+0.349	−0.051	+0.400	***
coach_challenge	98	−0.429	−0.898	+0.469	**

E10: Between-group recovery by fine subtype

subtype	n	μ	ctrl μ	Δ vs ctrl (sig)
tv_mandatory	1,486	+0.283	+0.371	−0.087 n.s.
stoppage	3,793	+0.125	+0.371	−0.246 ***
coach_absorb	1,563	+0.460	+0.371	+0.089 n.s.
coach_discretionary	4,694	+0.432	+0.371	+0.062 n.s.
mistagged_discretionary	11,543	+0.430	+0.371	+0.059 n.s.
coach_challenge	208	−0.596	+0.371	−0.967 ***

E11: Game phase and clutch conditioning

condition	endo n	endo μ	exo μ	ctrl μ	Δ endo-ctrl	Δ exo-ctrl
Q1 early (0–360s)	1,415	+0.408	+0.327	+0.470	−0.062 n.s.	−0.143 n.s.
Q1 late (360–720s)	1,231	+0.568	+0.322	+0.386	+0.182 n.s.	−0.063 n.s.
Q2 early (720–1080s)	2,451	+0.585	+0.303	+0.470	+0.114 n.s.	−0.167 n.s.
Q2 late (1080–1440s)	1,510	+0.291	+0.148	+0.283	+0.008 n.s.	−0.136 n.s.
Q3 early (1440–1800s)	2,517	+0.402	+0.289	+0.308	+0.094 n.s.	−0.018 n.s.
Q3 late (1800–2160s)	1,680	+0.327	+0.144	+0.407	−0.080 n.s.	−0.262 n.s.
Q4 early (2160–2520s)	2,706	+0.433	+0.089	+0.317	+0.116 n.s.	−0.228 n.s.
Q4 late (2520–2880s)	2,559	+0.406	+0.100	+0.272	+0.134 n.s.	−0.172 n.s.
Clutch (Q4 last 5min, $\lvert m\rvert \leq 5$)	993	+0.273	−0.253	+0.158	+0.115 n.s.	−0.411 n.s.
Non-clutch	15,244	+0.441	+0.249	+0.378	+0.063 n.s.	−0.129 *

E12: Team quality conditioning (NET_RATING)

condition	endo n	endo μ	exo μ	ctrl μ	Δ endo-ctrl	Δ exo-ctrl
Sufferer much worse ($\Delta \leq -5$)	4,097	−0.148	−0.134	−0.224	+0.076 n.s.	+0.090 n.s.
Sufferer worse ($-5 < \Delta \leq -1$)	3,822	+0.313	+0.013	+0.221	+0.092 n.s.	−0.208 n.s.
Evenly matched ($\lvert\Delta\rvert < 1$)	2,392	+0.656	+0.170	+0.341	+0.315 **	−0.170 n.s.
Sufferer better ($1 \leq \Delta < 5$)	3,328	+0.597	+0.453	+0.556	+0.041 n.s.	−0.103 n.s.
Sufferer much better ($\Delta \geq 5$)	2,598	+1.097	+0.924	+0.986	+0.111 n.s.	−0.062 n.s.

E13: Substitution-adjusted analysis

±30s subs	endo n	endo μ	exo μ	ctrl μ	Δ endo-ctrl	Δ exo-ctrl
0 subs	5,175	+0.426	+0.274	+0.489	−0.063 n.s.	−0.215 *
1–2 subs	3,864	+0.348	+0.160	+0.237	+0.110 n.s.	−0.077 n.s.
3+ subs	7,198	+0.479	+0.248	+0.262	+0.217 ***	−0.014 n.s.

E14: Running-team in-game PPP

condition	endo n	endo μ	exo μ	ctrl μ	Δ endo-ctrl	Δ exo-ctrl
Running PPP < 1.00	4,744	+0.343	+0.284	+0.145	+0.197 *	+0.139 n.s.
Running PPP 1.00–1.15	5,740	+0.429	+0.238	+0.341	+0.088 n.s.	−0.103 n.s.
Running PPP 1.15–1.30	4,158	+0.441	+0.255	+0.420	+0.021 n.s.	−0.164 n.s.
Running PPP ≥ 1.30	1,595	+0.672	+0.104	+0.474	+0.198 n.s.	−0.370 *